Biospectra : ISSN: 0973-7057, Vol. 17(2), September, 2022, pp. 303-306
An International Biannual Refereed Journal of Life Sciences

55(PS).
Plant Science

ISSN : 0973-7057

Int. Database Index: 616 www.mjl.clarivate.com

An annual algal diversity of Simaria Pond

Mrityunjay Kumar* & Radha Krishna Jha
University Department of Botany, Ranchi University, Ranchi, Jharkhand, India

Received : 23rd July, 2022 ; Revised : 25th August, 2022

Abstract- Algae has drawn much attention due to their primary productivity in the water food chain of water ecosystem diversity, their biological assessment of water quality, pollution abatement capacity and as a source of structurally novel and biologically active metabolite with antimicrobial capacity etc.

Key words: Simaria pond, Deoghar District, Jharkhand, Fresh water, Algal flora

INTRODUCTION

Algae are large group of prokaryotic and eukaryotic photosynthetic organism found in many different forms viz. Individual, colonies or extended filament and exhibit vast diversity.[1] They are cosmopolitan in nature, found everywhere like oceans, lakes, river, ponds, puddles most surface and fresh water etc.

Algae play an important role as primary produce of water ecosystem for various consumers of aquatic fauna and rich source of protein, carbohydrate and especially essential fatty acids.[2] They are one of the most helpful indicators of water quality due to their rapid response to environment changes related to large animals and plants. Thus, algae are used for biological assessment of water quality and bioindicator of eutrophication. The algae viz; planktonic and benthic are found on surface water and on depth respectively.

The dominance of green algae and Basillariophytes in relatively clean and oligotrophic water bodies, whereas blue green algae bloom formation indicates that the water body is polluted or eutrophic.[3]

There is great seasonal variation throughout year i.e., spring, summer, monsoon, winter. This variation has major impact on the water quality of the water body, changes in water quality parameters of water body influence the algal community of the water body nearby area.[4] Therefore, the present investigation has been carried out to observe the different algal community on monthly basis.

MATERIALS & METHODS

Study area

Simaria pond located in Deoghar District of Jharkhand. It is located near Simaria on the basis of literature survey, it appears that no systematic survey has been done in this area. So present work has been taken. Thus, the present study was made on the observation of algal flora in the Simaria pond on the monthly basis. Sampling was done from several fixed spots in the water body of the Simaria pond. The samples were collected at 30 days interval from the fixed spot. The samples were collected

*Corresponding author :
Phone : 9204481317
E-mail : sk750154@gmail.com

in a wide mouth glass bottle (500ml). Samples were preserved in 4% formalin solution deposited at University Department of Botany, Ranchi University Ranchi. For a detailed study samples were stained with iodine mounted in glycerine and examined under the different magnifications of microscope to identify the species. Species identification was done with the help of various key given by several authors.[5-8] Several other literatures were also studied for detailed description and identification.

RESULTS & DISCUSSION

The distribution of algae found in the Simaria Pond water body shows 53 species during one year study (2020-2021) out of these 31 cynophyceae, 14 chlorophyceae, 8 Bacillariophyceae and 1 euglenophyceae forms were observed. Such a vast quantitative variance in algal profile is amazing. Detailed examination shows that algae exhibit a high degree of qualitative variance. The occurrence and periodicity of algal species reported are given in Table 1.

Table 1- The occurrence and periodicity of algal species from Simaria Pond

SN	Algal Species	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
1	*Anabaena fertilissima*	+	+	+	-	-	-	-	-	+	+	+	+
2	*Anabaena iyengari*	+	+	+	+	-	-	-	-	+	+	+	+
3	*Aphanocapsa littoralis*	+	+	+	-	-	-	-	-	-	+	+	+
4	*Aphanothece microscopica*	+	+	+	+	+	-	-	-	-	-	+	+
5	*Calothrix gloeocola*	+	+	+	+	-	-	-	-	+	+	+	+
6	*Chlorella vulgaris*	+	+	+	+	-	-	-	-	+	+	+	+
7	*Chroococcus minor*	+	+	+	+	+	+	+	+	-	-	-	-
8	*Chroococcus minutes*	+	+	+	+	-	-	-	-	-	+	+	+
9	*Cladophora glomerarata*	+	+	+	-	-	-	-	+	+	+	+	+
10	*Clostrium venus*	-	-	-	-	-	-	-	-	+	+	+	+
11	*Cyclotella spp.*	+	+	+	-	-	+	+	-	-	+	+	+
12	*Cylindrospermum spp.*	+	+	+	+	-	-	-	-	-	+	+	+
13	*Euglena minuta*	+	+	+	+	-	-	-	-	-	-	+	+
14	*Hydrodictyon reticulatum* (L)	-	-	-	-	-	-	-	+	+	-	-	-
15	*Gleocapsa magma*	+	+	-	-	-	-	-	-	-	-	+	+
16	*Lyngbya contorta*	-	-	+	+	+	+	+	-	-	-	-	-
17	*Lyngbya epiphytica*	-	+	+	+	-	-	-	-	-	-	-	-
18	*Merismopedia glauca*	-	+	+	+	+	+	+	+	-	-	-	-
19	*Merismopedia tenuissima*	+	+	+	-	-	-	-	+	+	+	+	+
20	*Microcystis robusta*	+	+	+	+	+	+	+	+	+	+	+	+
21	*Microcystis viridis*	+	+	+	-	-	-	-	-	-	-	-	-
22	*Microcystis marginata*	+	+	+	-	-	-	-	+	+	+	-	+
23	*Mougeotia calcarean*	+	+	+	+	-	-	-	-	-	-	-	-
24	*Navicula major*	-	-	+	+	+	+	-	-	-	-	-	-
25	*Navicula microspore*	+	+	+	+	-	-	-	-	-	-	-	-
26	*Navicula viridis*	+	+	+	+	-	-	-	-	-	-	-	-
27	*Nostoc calcicole*	+	+	-	-	-	-	-	-	+	+	+	+
28	*Nostoc carneum*	+	+	-	-	-	-	-	-	+	+	+	+
29	*Oscillatoria limosa*	+	+	+	-	-	-	-	-	-	+	+	+
30	*Oscillatoria princeps*	+	+	-	-	-	-	-	-	-	-	+	+
31	*Oscillatoria tenuis*	+	-	-	-	-	-	-	-	-	+	+	+
32	*Oscillatoria annae*	+	+	+	-	-	-	-	-	+	+	+	+
33	*Pediastrum boryanum*	+	+	+	-	-	-	-	+	+	+	+	+
34	*Scenedesmus obliques*	+	+	+	-	-	-	+	-	-	+	+	-
35	*Scenedesmus dimorphus*	+	+	+	+	+	-	-	-	-	-	-	-
36	*Spirogyra regularis*	+	+	+	+	+	+	+	+	+	+	+	+
37	*Spirogyra submaxima*	+	+	+	+	+	+	+	+	+	+	+	+
38	*Spirulina gigantea*	+	+	+	+	-	-	-	-	-	-	+	+
39	*Spirulina major* Kutzing	+	+	+	+	-	-	-	-	-	-	+	+
40	*Ulothrix zonata*	+	+	-	-	-	-	+	+	+	+	+	+
41	*Ulothrix equalis*	+	+	-	-	-	-	+	+	+	+	+	+
42	*Zygnema collimsianum*	+	+	+	-	-	-	+	+	+	+	+	+

CONCLUSION

The algal community of the study area is dominated by cyanophyceae member at the same time certain algae which contribute in degradation of water quality by showing toxic effect (e.g. species of *Anabaena, Lyngbya* and *Microcystis*), allergenic diseases viz. Species of *Chlorella, Scendesmus, Anabaena, Microcystis* and *Oscillatoria*. Colour (Species of *Chlorella, Microcystis*) taste and odour (Species of *Chlorella, Cladophora, Hydrodictyon, Scendesmus, Spirogyra, Fragillaria, Anabaena, Nostoc, Oscillatoria, Euglena* were reported from study area.

Thus, the study side can be considered to be eutrophic with the mesosaprobic condition and requires management strategies to improve the water quality. As the study is the very first report of algal diversity from the study side. the extracted information from this study can be used as baseline data for climate change and global warming studies. Further after a sufficient interval of time, one can assess the impact of changing environment on the algal diversity from the same study area.

AKNOWLEDGEMENT

I am thankful to the Head, University Department of Botany for constant encouragement and providing the necessary Lab facilities.

REFERENCES

1. **Mohan J., Narayan S., Kumar H. and Mohan N. 2007**. Diversity of blue green algae in Allen Forest Lake. *Indian Hydrobiology.* **10(1):**123-127.
2. **Mishra P. K., Prakash J., Srivastava A. K. and Singh P. K. 2004.** Some fresh water planktonic algae from Sant Kabir Nagar, Uttar Pradesh. *Phytotaxonomy*. **4:** 87-94.
3. **Zaidi J., Ganesh S. and Pal A. 2011.** Assessment of water quality in river Betwa and Pahuj around Jhansi city, UP. *International Journal of current Research*. **33(5):**27-31.
4. **Verma P. U., Chandawat D. K. and Solanki H. A. 2011.** Seasonal variation in physico- chemical and phytoplankton analysis of Karnataka Lake. *Life Science Leaflets*. **19:**842-854.
5. **Anand N. 1998.** Indian Freshwater Microalage. Bishen Singh Mahendra Pal Singh Publicartion, Dehradun, India.
6. **Desikachary T. V. 1959**. Cyanophyta, Monograph on blue green algae. I.C.A.R., New Delhi, India, pp. 1- 689.
7. **Prescott G. W. 1964.** The freshwater algae. Brown Company Publisher, Dubuque, Lowa.
8. **Philipose M. T. 1967.** Chlrococcales, Monograph on algae ICAR, New Delhi.

ADDITIONAL REFERENCES

9. **Bajpai O., Mishra S., Mohan N. Mohan J. and Gupta R. K. 2013.** Physico chemical characteristics of Lakhna Devi temple water tank, Lakhna, Bakewar, Etawaha, U.P. with reference to Cynobacterial Diversity. *International journal of Environment.* **1(1):** 20-28
10. **Chacko P. I. and Krishnamurthy K. 1954.** On the plankton of three freshwater ponds in Madras. Symposium on marine and fresh water plankton. *Indo-Pacific Council Bangkok*. Pp. 103-107.
11. **Ganpati S. V. and Chako P. I. 1951.** Investigation on plankton production and limnology of our fish ponds in Madras. *Proceedings of the International Association of Theoretical & Applied Limnology*. **11:**111-112.
12. **Harris J. M. and Vinobaba P. 2012.** Impact of water quality on Species Composition and Seasonal Fluctuation of Planktons of Batticaloa lagoon, Sri Lanka. *Journal of Ecosystem & Ecography*. 2: 117
13. **Kumar P., Wanganeo A., Sonaullah F. and Wanganeradeso R. 2012.** Limnological Study on two High Altitude Himalayan Ponds, Badrinath, Uttarakhand. *International Journal of Ecosystem.* **2(5):** 103-111.
14. **Rajagopal T., Thangamani A. and Archunan G. 2010.** Comparison of physico-chemical parameters and phytoplankton species diversity of two perennial ponds in Sattur area Tamil Nadu. *Journal of Environmental Biology*. **31(5):**787-794.
15. **Singh U. B., Ahluwalia A. S., Sharma C., Jindal R. and Thakur R. K. 2013.** Planktonic indicators: A promising tool for monitoring water quality (early warning signals). *Ecology, Environment and Conservation.* **19(3):**793-800.

16. Srivastava N., Sussela M. R. and Toppo K. 2014. Fresh water Cynobacteria of Sai River near Lucknow. Uttar Pradesh. *Tropical Plant Research.* **1(2):**11-16.

17. Talpasayi E. R. S. 1962. The Myxophyceae of the Kumaon Hills U.P. India, *Proceeding of Indian Academy of Sci.* **55:**251-255.

18. Tiffany L. H. & Britton M. E. 1952. Monograph on the algae of Illinoids, the University of Chicago Press,1952.

***